FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. BD

Sol. Length $\propto G^x c^y h^z$

$L = \left[M^{-1}L^3T^{-2}\right]^x \left[LT^{-1}\right]^y \left[ML^2T^{-1}\right]^Z$

By comparing the power of M, L and T in both sides we get

$-x + z = 0, 3x + y + 2z = 1$ and $-2x - y - z = 0$

By solving above three equations we get

$x = \frac{1}{2}, y = -\frac{3}{2}, z = \frac{1}{2}$

2. BCD

Sol. Sign between *x* and *t* is $\oplus$ hence wave is propagating in negative *x* direction.

$$\text{wave velocity} = \frac{\text{coefficient of t}}{\text{coefficient of x}} = \frac{3}{1}$$

Max value of *y* is amplitude, *y* has maximum value when denominator is least i.e.

when $(x + 3t)^2$ is zero

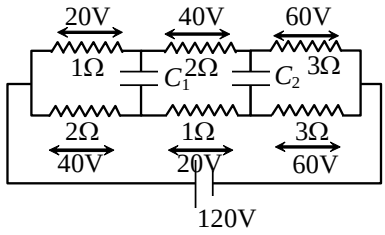
$y = 3m$

shape of pulse is given by $y = \frac{6}{2 + x^2}$ $(t = 0)$

shape of pulse is symmetrical out y-axis

3. BD

Sol. Now, potential difference across C_1 is 20 V and across C_2 is zero.
$\therefore$ charge stored in C_1 is 40 μC and in C_2 is zero.



4. BC

Sol. $v^2 = a - bx^2$

differentiating both side w.r.t. time

$2v\dfrac{dv}{dt} = -2bx\left(\dfrac{dx}{dt}\right)$ $\qquad \dfrac{dv}{dt} = A$ (acceleration)

$A = -bx$

Hence motion will be S.H.M. $\omega^2 = b$, $\qquad 4\pi^2 f^2 = b$, $\qquad f = \dfrac{\sqrt{b}}{2\pi}$

$T.E. = \dfrac{1}{2}m\omega^2 (\text{Amplitude})^2 = \dfrac{1}{2}m\,b\,\dfrac{a}{b} = \dfrac{1}{2}ma$

5. BD

Sol. Stopping potential does not depend on intensity and saturation current depends on intensity. Intensity I due to point source $\propto \dfrac{1}{r^2}$ where r is distance from the source.

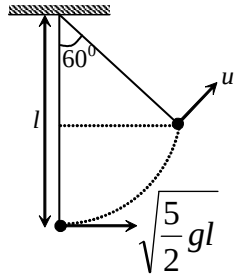
6. ABCD

Sol. $T - mg\cos 60^0 = \dfrac{mv^2}{l}$ … (i)

$\dfrac{1}{2}m\left(\sqrt{\dfrac{5}{2}gl}\right)^2 - mg\dfrac{l}{2} = \dfrac{1}{2}mv^2$ … (ii)

$v = \sqrt{\dfrac{3}{2}gl}$ and $T = 2mg$

Net force is in horizontal direction when $\theta = 60^0$



7. ABCD

Sol. $\vec{F} = 3\hat{i} - 4\hat{j}$, $W_{con.f} = U_i - U_f = 34 - 9 = 25$ & $\dfrac{1}{2}mv^2 - 16 = 34$

8. B

Sol. $Q_{outside}$ = earthed plate will have zero charge on outer surface for its zero potential condition.

9. B

Sol. Earthed plate will have zero charge on outer surface for its zero potential condition.

10. A

Sol. $Q_{outside} = (Q_1+Q_2)/2$, and conservation of charge.

11. C

Sol. $P = \frac{\rho RT}{M}$

12. B

Sol. VT = constant

$\Rightarrow$ PV^2 = constant

Thus, $\Delta W = \frac{nR(T_2 - T_1)}{1-2}$

13. D

Sol. $\Delta U = 0$

$\Rightarrow$ $\Delta T = 0$, process is isothermal.

SECTION – B

14. 7

Sol. Given, voltage V = (100 ± 5)V

Current I = (10 ± 0.2)A

According to Ohm's law, V = IR or R = V/I

Taking log of both sides,

Log R = log V – log I

Differentiating, we get

$\frac{\Delta R}{R} = \frac{\Delta V}{V} - \frac{\Delta I}{I}$

For maximum error, $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$

Multiplying both sides by 100 for taking percentage, we get

$\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$

Percentage error in resistance R

$= \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$

$= \frac{5}{100} \times 100 + \frac{0.2}{10} \times 100$ = 7%.

15. 5

Sol. $5 = \frac{i}{2}(6) + i(2) + iR$

$\Rightarrow$ $5 = 3i + 2i + iR$

$\Rightarrow$ $i = \frac{5}{5+R}$

$P = i^2R = \left(\frac{5}{5+R}\right)^2 R$

$P = i^2R$

$$\frac{dP}{dR}=0$$
$$R=5\Omega.$$

16. 3

Sol. $PT^2 = k$

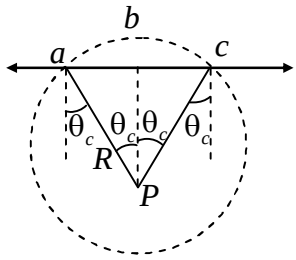
$$\gamma=\frac{1}{V}\left(\frac{dV}{dT}\right) \qquad \ldots(i)$$
$$\left(\frac{nRT}{V}\right)T^2=k$$
$$\Rightarrow \quad \frac{T^3}{V}=k'$$
$$\Rightarrow \quad V=\frac{T^3}{k'} \qquad \ldots(ii)$$
$$\frac{dV}{dT}=\frac{3T^2}{K'} \qquad \ldots(iii)$$

From (i), (ii) and (iii)

$$\gamma=\left(\frac{k'}{T^3}\right)\times\left(\frac{3T^2}{k'}\right)=\frac{3}{T}$$

17. 1

Sol. The light escape is confined within a cone of apex angle '$2\theta_c$' where θ_c is the critical angle. Imagine a sphere with source of light as its centre and the surface area abc is A.



here $A=\int_0^{\theta_c} 2\pi R^2\sin\theta\, d\theta = 2\pi R^2(1-\cos\theta_c)$

$$=\pi R^2 \qquad \left[\therefore \theta_c=\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=60^o\right]$$

$$\therefore \text{ Power transfer } = P\times\frac{A}{4\pi R^2}=4\times\frac{1}{4}=1W$$

18. 2

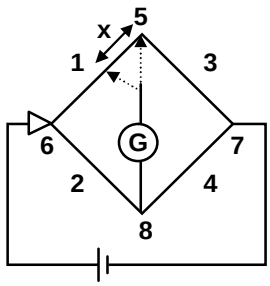
Sol. For getting null point

$$\frac{R_1}{R_3}=\frac{R_2}{R_4}$$

as R2 = R4

R1 = R3

Let the pointer at point 5 is moved to left by distance x to get nul point as shown in the figure. If resistance per unit length of wire 3 is r then that of wire 1 will be 2r.



$(8 - x)2r = x \times 2r + 8 \times r$

$4x = 8$

$x = 2m$

Chemistry

PART – II

SECTION – A

19. AD
Sol. The n-factor of $(NH_4)_2Cr_2O_7$ is 6.

20. ABD
Sol. They contain weak acid and salt of strong base.

21. AB
Sol. The overall order of the reaction is 2.

22. AB
Sol. H_2 and He are lighter than CH_4.

23. AD
Sol. $2NaI + 2H_2SO_4 \longrightarrow Na_2SO_4 + SO_2 + I_2 + 2H_2O$
KBr reacts in the same way.

24. CD
Sol. Ni^{2+} contains eight and Zn^{2+} contains 10 electros.
$\therefore$ $Ni^{2+} = t_{2g}^6 e_g^2$ and $Zn^{2+} = t_{2g}^6 e_g^4$

25. ACD
Sol. It has three geometrical isomers.

26. C
Sol. Coordination number of $Na^+ = Cl^- = 6$. It has fcc lattice and Na^+ occupies the octahedral voids.

27. C
Sol. CsCl has b.c.c unit cell. The coordination number of Cs^+ is 8.

28. D
Sol. Simple cubic is a primitive unit cell. It's coordination number is 6.

29. B
Sol. $2\ HCHO \xrightarrow{NaOH} HCOONa + CH_3OH$

30. C
Sol. $C_2H_5NH_2 + HNO_2 \longrightarrow C_2H_5OH + N_2 + H_2O$

31. C
Sol. $CH_3COOH + C_2H_5OH \xrightarrow{H^+} CH_3COOC_2H_5 + H_2O$.

SECTION – B

32. 4

Sol. $\frac{r_x}{r_{SO_2}} = \sqrt{\frac{M_{SO_2}}{M_X}} \Rightarrow \frac{4r_{SO_2}}{r_{SO_2}} = \sqrt{\frac{64}{M_X}}$

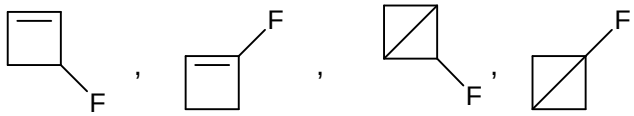
$\therefore M_X = 4$

33. 6

Sol. $K_{sp} = S^2 \Rightarrow S = \sqrt{16 \times 10^{-4}} = 4 \times 10^{-2}$

$= x \times 10^{-y} \Rightarrow x + y = 4 + 2 = 6$

34. 4

Sol. The isomers are 

35. 3

Sol. $C_2H_5NH_2 \xrightarrow{CH_3I} C_2H_5NHCH_3 \xrightarrow{CH_3I} C_2H_5N(CH_3)_2 \xrightarrow{CH_3I} (C_2H_5)N^+(CH_3)_2I^-$

36. 4

Sol. The unit cell F.C.C is formed by S^{2-} ions. Zn^{2+} occupies half of the tetrahedral voids.

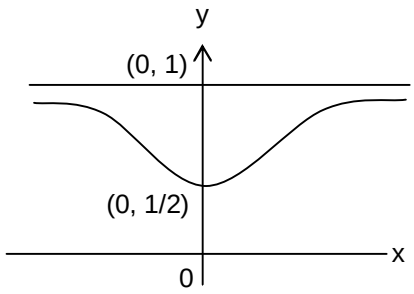
Mathematics

PART – III

SECTION – A

37. BCD

Sol. Domain is $x \in R$



Also $f(x) = \left[\cos\left(\tan^{-1}(\sin\theta)\right)\right]^2$

where $\cot\theta = x$

$$= \left[\cos\left(\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)\right]^2$$

$$= (\cos\phi)^2 \text{ where } \tan\phi = \frac{1}{\sqrt{1+x^2}} = \left(\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right)^2$$

$$f(x) = \frac{1+x^2}{2+x^2} = 1 - \frac{1}{2+x^2}$$

Range is $\left[\frac{1}{2}, 1\right)$; $f'(x) = \frac{2x}{(2+x^2)^2}$

Hence $f'(0) = 0$

Also $\lim\limits_{x\to\infty} f(x) = 1$

Hence (B), (C), (D)

38. ABCD

Sol. We have $\alpha(\hat{i} - 3\hat{k}) + \beta(2\hat{i} + \hat{j}) + \gamma(\hat{i} - \hat{j} + \hat{k}) = -2\hat{i} - 5\hat{j} + 6\hat{k}$

On comparing, we get

$\alpha + 2\beta + \gamma = -2$ ……(i)

$\beta - \gamma = -5$ ……(ii)

$-3\alpha + \gamma = 6$ ……(iii)

Solving the equations (i), (ii) and (iii), we get

$\alpha = -1, \beta = -2, \gamma = 3$

39. CD

Sol. (α, α^2) lie on the parabola $y = x^2$

(α, α^2) must lie between the asymptotes on hyperbola

$\frac{x^2}{1} - \frac{y^2}{4} = 1$ in 1st and 2nd quadrant.

$\therefore$ Asymptotes are $y = \pm 2x$.

$\Rightarrow \quad 2\alpha < \alpha^2$

$\Rightarrow \quad \alpha < 0$ or $\alpha > 2$

and $\quad -2\alpha < \alpha^2$

$\alpha < -2$ or $\alpha > 0$

$\therefore \ \alpha \in (-\infty, -2)$ or $\alpha \in (2, \infty)$

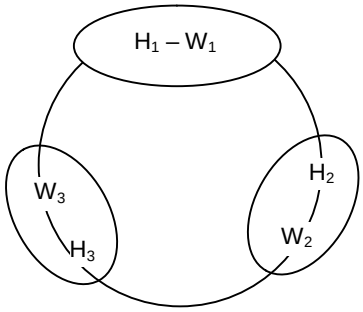
40. BC

Sol. n (All couples are together)

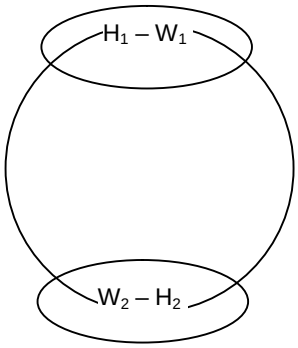
$= 2 \times 2 \times 2 \times 2 = 16$ ways

$\therefore$ x = Total – 16

$x = 5! - 16 = 104$



To find y, first Let H_1, W_1, H_2, W_2 sit in $2 \times 2 = 4$ ways



Now H_3, and W_3 can sit in 2 ways

$\therefore y = 4 \times 2 = 8$

41. A

Sol. $\left(9 + 4\sqrt{5}\right)^n = I + f$

Let $\left(9 - 4\sqrt{5}\right)^n = f'$

$I + f + f' = \left(9 + 4\sqrt{5}\right)^n + \left(9 - 4\sqrt{5}\right)^n$

= Even integer

$0 < f + f' < 2$

$\therefore$ I is an odd integer

$\Rightarrow (a)$

42. ABCD

Sol. Now, applying $C_3 \to C_3 - g(\theta)C_1 - f(\theta)C_2$

$$\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha+\theta) \\ f(\beta) & g(\beta) & f(\beta+\theta) \\ f(\gamma) & g(\gamma) & f(\gamma+\theta) \end{vmatrix} = 0$$

$\left[\because f(x+y) = f(x)g(y) + g(x)f(y)\right]$

43. ABC

Sol. The conditions for the existence of a solution are

1. $2\cos 2x - 1 \neq 0 \Rightarrow \cos 2x \neq \frac{1}{2} \Rightarrow 2x \neq \frac{\pi}{6}$

2. $\tan x \neq 0 \Rightarrow x \neq n\pi$

3. $\cos^2 x - 3\sin^2 x \neq 0 \Rightarrow \tan^2 x \neq \frac{1}{3} \Rightarrow x \neq \pm\frac{\pi}{6}$

As $2\cos x - 1 = 2\left(\cos^2 x - \sin^2 x\right) - 1$

$= 2\left(\cos^2 x - \sin^2 x\right) - \left(\cos^2 x + \sin^2 x\right)$

$= \cos^2 x - 3\sin^2 x$

So the given equation can be written as $b\sin x = b + \sin x$

$\Rightarrow \quad \sin x = \frac{b}{b-1}$

Which is possible if $-1 \leq \frac{b}{b-1} \leq 1$

$\Rightarrow \quad \frac{2b-1}{b-1} \geq 0$ and $\frac{1}{b-1} \leq 0$

$\Rightarrow \quad 2b - 1 \leq 0 \Rightarrow b \leq \frac{1}{2}$

But for $b = \frac{1}{2}, \sin x = -1$ which is not possible as $x \neq -\frac{\pi}{2}$.

Hence equation has a solution if $b < \frac{1}{2}$.

44. B

45. B

46. B

Sol. (for Q. 44 – 46)

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0 \Rightarrow \alpha = 4$$

Put $z = 0$ in P_1 and P_2

$2x + y = 1$ and $x - y = 2$

$\therefore \quad P \equiv (1, -1, 0)$

Put x = 0 in P_1 and P_2

$y + z = 1$ and $-y + z = 2$

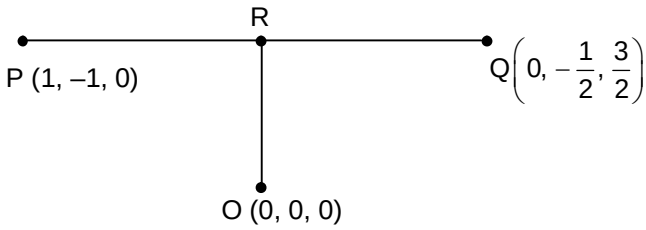
$\therefore \ Q \equiv \left[0, -\frac{1}{2}, \frac{3}{2}\right]$

$\vec{PQ} = -\hat{i} + \frac{1}{2}\hat{j} + \frac{3}{2}\hat{k}$

Projection of PQ on x – axis $= \left|\vec{PQ}.\hat{i}\right| = 1$

Equation of line PQ is

$$\frac{x-1}{-1}=\frac{y+1}{1/2}=\frac{z}{3/2}=\lambda$$



Let $R\equiv\left(-\lambda+1,\frac{\lambda}{2}-1,\frac{3\lambda}{2}\right)$

$\vec{OR}.\vec{PQ}=0$

$\Rightarrow -1(-\lambda+1)+\frac{1}{2}\left(\frac{\lambda}{2}-1\right)+\frac{9\lambda}{4}=0$

$\Rightarrow \quad \lambda=\frac{3}{7}$

$\therefore \quad R\equiv\left(\frac{4}{7},-\frac{11}{14},\frac{9}{14}\right)$

$\Rightarrow \quad 7a+14b+14c=2$

Area $=\frac{1}{2}\left[\vec{PQ}\times\vec{OQ}\right]=\sqrt{\frac{19}{16}}$

$\therefore \quad a-b=3$

47. C
48. A
49. B

Sol. (for Q. 47 – 49)

$$A^{-1}=\begin{bmatrix}1&0&0\\-2&1&0\\1&-2&1\end{bmatrix}$$

$$X_1=A^{-1}\begin{bmatrix}1\\0\\0\end{bmatrix}=\begin{bmatrix}1\\-2\\1\end{bmatrix},\ X_2=A^{-1}\begin{bmatrix}2\\3\\0\end{bmatrix}=\begin{bmatrix}2\\-1\\-4\end{bmatrix}$$

$$X_3=A^{-1}\begin{bmatrix}2\\3\\1\end{bmatrix}=\begin{bmatrix}2\\-1\\-3\end{bmatrix}$$

$$\therefore \quad X=\begin{bmatrix}1&2&2\\-2&-1&-1\\1&-4&-3\end{bmatrix}$$

$\therefore \quad \det X=3$

$\det A=1$

$\det(\text{adj}\,X)=3^2=9$

Sum of the elements of X^{-1} is zero because $X^{-1}=\frac{1}{3}\begin{bmatrix}-1&-2&0\\-7&-5&-3\\9&6&3\end{bmatrix}$

SECTION – B

50. 3

Sol. $\alpha+\beta+\gamma=c$

$\therefore \quad \alpha\beta+\beta\gamma+\gamma\alpha=4;\alpha\beta\gamma=2$

$\therefore \quad \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}=\frac{4}{2}=2$

$\because \quad f\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)=2$

$\Rightarrow \quad f(2)=2$

$\Rightarrow \quad 8-4c+8-2=2$

$\Rightarrow \quad c=3$

51. 3

Sol. Since maximum value of $\cos^{-1}x$ in $[0,1]=\frac{\pi}{2},\sum_{r=1}^{k}\cos^{-1}\beta_r=\frac{k\pi}{2}$ is possible if and only if each $\cos^{-1}\beta_r=\frac{\pi}{2}\Leftrightarrow\beta_r=0$.

$A=\sum_{r=1}^{k}(\beta_r)^r=0$

Thus, $\lim_{x\to0}\frac{(1+x)^{1/3}-(1-2x)^{1/4}}{x+x^2}$

$=\lim_{x\to0}\frac{\left(1+\frac{x^2}{3}+0(x^4)\right)-\left(1-\frac{x}{2}+0(x^2)\right)}{x(1+x)}$

$=\lim_{x\to0}\frac{\frac{1}{2}+\frac{x}{3}+0(x^2)}{1+x}=\frac{1}{2}$. Thus the required limit is 249.

52. 7

Sol. Position vector of orthocenter is H $\left(a\hat{i}+b\hat{j}+c\hat{k}\right)$

$\overrightarrow{AH}\perp\overrightarrow{BC}\Rightarrow\left((a-1)\hat{i}+b\hat{j}+c\hat{k}\right).\left(-2\hat{j}+3\hat{k}\right)=0$

$\Rightarrow -2b+3c=0$

$\overrightarrow{BH}\perp\overrightarrow{AC}\Rightarrow\left(a\hat{i}+(b-2)\hat{j}+c\hat{k}\right).\left(-\hat{i}+3\hat{k}\right)=0$

$\Rightarrow -a+3c=0$ and $\overrightarrow{CH}\perp\overrightarrow{AB}\Rightarrow\left(a\hat{i}+b\hat{j}+(c-3)\hat{k}\right)\left(-\hat{i}+2\hat{j}\right)=0$

$\Rightarrow -a+2b=0\Rightarrow a=2b=3c=\lambda$ (say)

$\Rightarrow H=\left(\lambda\hat{i}+\frac{\lambda}{2}\hat{j}+\frac{\lambda}{3}\hat{k}\right)$

H lies on plane ABC

So, points A, B, C and H are collinear.

$$\therefore \begin{vmatrix} 1-0 & 0-2 & 0-0 \\ 1-0 & 0-0 & 0-3 \\ 1-\lambda & 0-\frac{\lambda}{2} & 0-\frac{\lambda}{3} \end{vmatrix} = 0$$

$$\Rightarrow 6 - 6\lambda - \frac{2\lambda}{3} - \frac{3\lambda}{2} = 0 \Rightarrow \lambda = \frac{36}{49}$$

Hence, position vector of orthocenter is $\left(\frac{36}{49}\right)\left(i + \frac{1}{2}\hat{j} + \frac{1}{3}\hat{k}\right)$.

$$\therefore \sqrt{\frac{6}{a-b-c}} = \sqrt{\frac{6}{\frac{36}{49} - \frac{18}{49} - \frac{12}{49}}} = 7$$

53. 7

Sol. Let $P(a\cos\theta_1, b\sin\theta_1), Q(a\cos\theta_2, b\sin\theta_2)$ and $R(a\cos\theta_3, b\sin\theta_3)$ be the vertices of the triangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The points on the auxiliary circle corresponding to these points are $P'(a\cos\theta_1, a\sin\theta_1)$, $Q'(a\cos\theta_2, a\sin\theta_2)$ and $R'(a\cos\theta_3, a\sin\theta_3)$

$\therefore \Delta_1 =$ Area of ΔPQR

$$= \frac{1}{2}\begin{vmatrix} a\cos\theta_1 & b\sin\theta_1 & 1 \\ a\cos\theta_2 & b\sin\theta_2 & 1 \\ a\cos\theta_3 & b\sin\theta_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}ab\begin{vmatrix} \cos\theta_1 & \sin\theta_1 & 1 \\ \cos\theta_2 & \sin\theta_2 & 1 \\ \cos\theta_3 & \sin\theta_3 & 1 \end{vmatrix}$$ and $\Delta_2 =$ Area of $\Delta P'Q'R'$

$$= \frac{1}{2}\begin{vmatrix} a\cos\theta_1 & a\sin\theta_1 & 1 \\ a\cos\theta_2 & a\sin\theta_2 & 1 \\ a\cos\theta_3 & a\sin\theta_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}a^2\begin{vmatrix} \cos\theta_1 & \sin\theta_1 & 1 \\ \cos\theta_2 & \sin\theta_2 & 1 \\ \cos\theta_3 & \sin\theta_3 & 1 \end{vmatrix}$$

Clearly, $\frac{\Delta_1}{\Delta_2} = \frac{1}{7} \Rightarrow \frac{\Delta_2}{\Delta_1} = 7$

54. 9

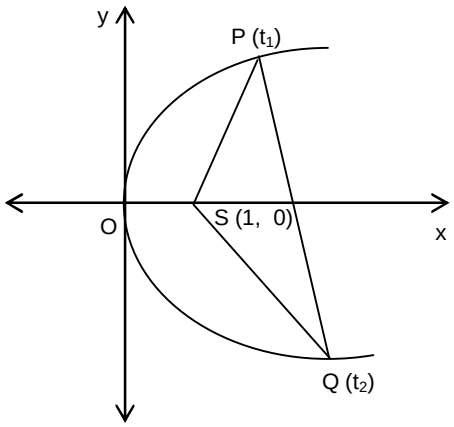
Sol. We have $3h = \left(t_1^2 + t_2^2 + 1\right)$ .....(1)

$3k = 2\left(t_1 + t_2\right)$ .....(2)

and $t_2 = -t_1 - \frac{2}{t_1}$ .....(3)

$\Rightarrow t_1 + t_2 = \dfrac{-2}{t_1}$

Eliminate t_1 and t_2 to get



$\therefore \quad \dfrac{3k}{2} = \dfrac{-2}{t_1} \Rightarrow t_1 = \dfrac{-4}{3k}$

$t_2 = \dfrac{3k}{2} + \dfrac{4}{3k}$

Substitute the values of t_1 and t_2 in (1) we get

$3h = \dfrac{16}{9k^2} + \left(\dfrac{3k}{2} + \dfrac{4}{3k}\right)^2 + 1$

$3h = \dfrac{32}{9k^2} + \dfrac{9k^2}{4} + 5$

$(3x - 5)36y^2 = 128 + 81y^4$

Hence $g = 36;\ k = 81$ and $w = 128$